

- Concentrate on your own journey
- We need to build a supportive atmosphere

Solving equations $Ax=b$ $\left\{ \begin{array}{l} A \text{ } r \times c \text{ matrix} \\ x \text{ unknown vector in } \mathbb{R}^c \\ b \text{ vector in } \mathbb{R}^r \end{array} \right.$

- It is convenient to do row operations to put $[A|b]$ (augmented matrix) in row echelon form
- Even more convenient to put it in reduced echelon form

Elementary row operations

- 1) Switching two rows
- 2) Scaling a row by a nonzero number
- 3) Adding a multiple of one row to another

- These operations are reversible
- Doing these ops to $[A|b]$ does not change the solⁿ set
- (Ex) "Doing an elementary row operation to a matrix M is the same as multiplying M on the left by a suitable matrix" Figure out the "suitable matrices" i.e.

$M \xrightarrow[\text{of type ① ② or ③}]{\text{operation}} EM$

Figure out what E should be for types ①, ②, ③

Row echelon form for a matrix M

$M \xrightarrow[\text{of row ops}]{\text{sequence}} \text{Matrix satisfying the following properties}$

- Defⁿ The first nonzero entry in any row is called a pivot
- Pivots in successive rows should move strictly to the right (so all entries below a pivot must be 0) and all rows without pivots (i.e. row consisting of only zeroes) must come at the bottom
- Customary to scale rows so that all pivots are = 1

For reduced echelon form (RREF or REF) we also require all entries above a pivot to be zero.

- [Fact/exercise • RREF of a matrix is unique.]
- Number of pivots in any echelon form is the same.

Example of RREF

$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8$

Multiple things are discussed here. I'll state them separately tomorrow

Given system is consistent iff $q_5, q_6 = 0$
 In that case gen solution is given by
 x_1, x_3, x_6, x_7 : arbitrary
 x_2, x_4, x_5, x_8 : expressed uniquely in terms of free vars

$$\begin{array}{cccccccc|c} 0 & 1 & a & 0 & 0 & b & c & 0 & q_1 \\ 0 & 0 & 0 & 1 & 0 & d & e & 0 & q_2 \\ 0 & 0 & 0 & 0 & 1 & f & g & 0 & q_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & q_4 \\ \hline & & & & 0 & & & & q_5 \\ \hline & & & & 0 & & & & q_6 \end{array}$$

Each $q_i =$ some expression in terms of original constants p_i on RHS of $Ax = \bar{p}$

Use of RREF in writing solutions to $Ax = b$

$$\begin{array}{l} x + 2y = b_1 \\ 2x + 5y = b_2 \end{array} \quad \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 2 & 5 & b_2 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 1 & b_2 - 2b_1 \end{array} \right]$$

Answer $x = 5b_1 - 2b_2$
 $y = b_2 - 2b_1$

$$\xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 5b_1 - 2b_2 \\ 0 & 1 & b_2 - 2b_1 \end{array} \right]$$

$$\begin{aligned} x+2y &= b_1 \\ 2x+4y &= b_2 \end{aligned} \quad \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 2 & 4 & b_2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 0 & b_2 - 2b_1 \end{array} \right]$$

For this system to have a solution, it is necessary (and sufficient) that $b_2 - 2b_1 = 0$

• When is $Ax=b$ consistent?

$$\left[\begin{array}{c|c} A & \begin{matrix} b_1 \\ \vdots \\ b_r \end{matrix} \end{array} \right] \rightsquigarrow \left[\begin{array}{c|c} \text{Echelon form for } A & \begin{matrix} \text{some linear} \\ \text{expressions in} \\ \text{terms of } b_1, \dots, b_r \end{matrix} \end{array} \right]$$

Answer $Ax=b$ is inconsistent precisely when there is a pivot in the last column of an echelon form of the augmented matrix $[A|b]$

(i.e. $Ax=b$ is consistent precisely when the following happens: if there are any zero rows in an echelon form of A , then in any echelon form of $[A|b]$ the last entries in those rows must be zero)

Q- When does the system $Ax=b$ have a solution regardless of which vector b is given to me?
[A is fixed, b is arbitrary]

$$\left[A | b \right] \xrightarrow{\text{row op}^n} \left[A' | b' \right]$$

↓
echelon form

Observe:

As b ranges over all possible vectors in $\mathbb{R}^r$,

b' also range over all ---//--- : just work

backward, from a desired b' to get a b that will lead to it (the desired b')

Ans to the question: $Ax = b$ is consistent for each b

- $\Leftrightarrow$ Echelon form of A has no zero rows
- $\Leftrightarrow$ Each row of $\text{REF}(A)$ has a pivot
- $\Leftrightarrow$ # pivots in $\text{REF}(A) = \#$ of rows r

We observed so far about $Ax = b$

- The system may or may not be consistent
- If it is consistent, then "dim" of solⁿ space = # of free variables b/c these variables can be given arbitrary values & then the rest of the variables ("pivot variables") are uniquely decided in terms of free variables.

• We do not have to worry about consistency if $b = \bar{0}$!

In this case we have $A\bar{x} = \bar{0}$. This system always has a solution, namely $\bar{x} = \bar{0}$.

$Ax = 0$ called a homogeneous system.

Discussion after class: will be done in class tomorrow

Take some $Ax = b$.

Importance of $Ax = 0$

- Let z be a solution. i.e. $Az = b$

Set of all solutions to $Ax=b$

$$= \{z + \text{each solution to } Ax=0\}$$

Reason: following simple calculation

$$\begin{array}{l|l} Ay=0 & A(w-z)=0 \\ \Rightarrow A(y+z)=b & \uparrow \\ & Aw=b \end{array}$$

• Set of solutions to $Ax=0$

$\equiv$ Null space of matrix A

$\equiv$ kernel/null space of map $(x \mapsto Ax)$

Definitions

$$\left\{ \begin{array}{l} \mathbb{R}^c \rightarrow \mathbb{R}^r \\ x \mapsto Ax \end{array} \right. \quad A: r \times c$$

• Study of function defined by

is going to be essentially linear algebra

(ex) Image of this map } are both vector spaces!
kernel of this map }